# Diffraction and Huygens' Principle

> **SYNOPSIS**
>
> Waves bend around corners. Huygens' principle says why, in one sentence, and that sentence then does something strange: it makes a **single** slit produce bright and dark bands — with the roles of $m\lambda$ and $(m + \tfrac{1}{2})\lambda$ **exactly reversed** from Note 10.

**BEFORE YOU START — PREDICT**

In Note 10, $d \sin\theta = m\lambda$ gave a **bright** fringe. In this note, $a \sin\theta = m\lambda$ gives a **dark** one. Same-looking equation, opposite answer. What is different about the physical situation? Guess now; Figure 2 settles it.

**PERFORMANCE EXPECTATIONS**

1. **State** Huygens' principle and use it to explain why a wave bends around an obstacle.
2. **Explain** the pairwise-cancellation argument for the single-slit minima.
3. **Apply** $a \sin\theta = m\lambda$ to locate the **dark** fringes of a single slit, and say why it is not the bright ones.
4. **Use** $I = I_0(\sin\beta/\beta)^2$ with $\beta = \pi D \sin\theta/\lambda$ to find the intensity at a given angle.

## 2. Diffraction

Waves **bend around a corner**. Ocean swell arriving at the end of a breakwater does not stop at the line of the wall and leave flat water behind it — the crests curl round into the sheltered area. That bending is **diffraction**.

### Huygens' principle

**HUYGENS' PRINCIPLE**

Every point of an advancing wavefront is a new centre of disturbance
from which emanate independent waves in all directions

Each point serves as a source of **secondary wavelets**, and the new wavefront is the **tangential surface** of all those secondary wavelets.

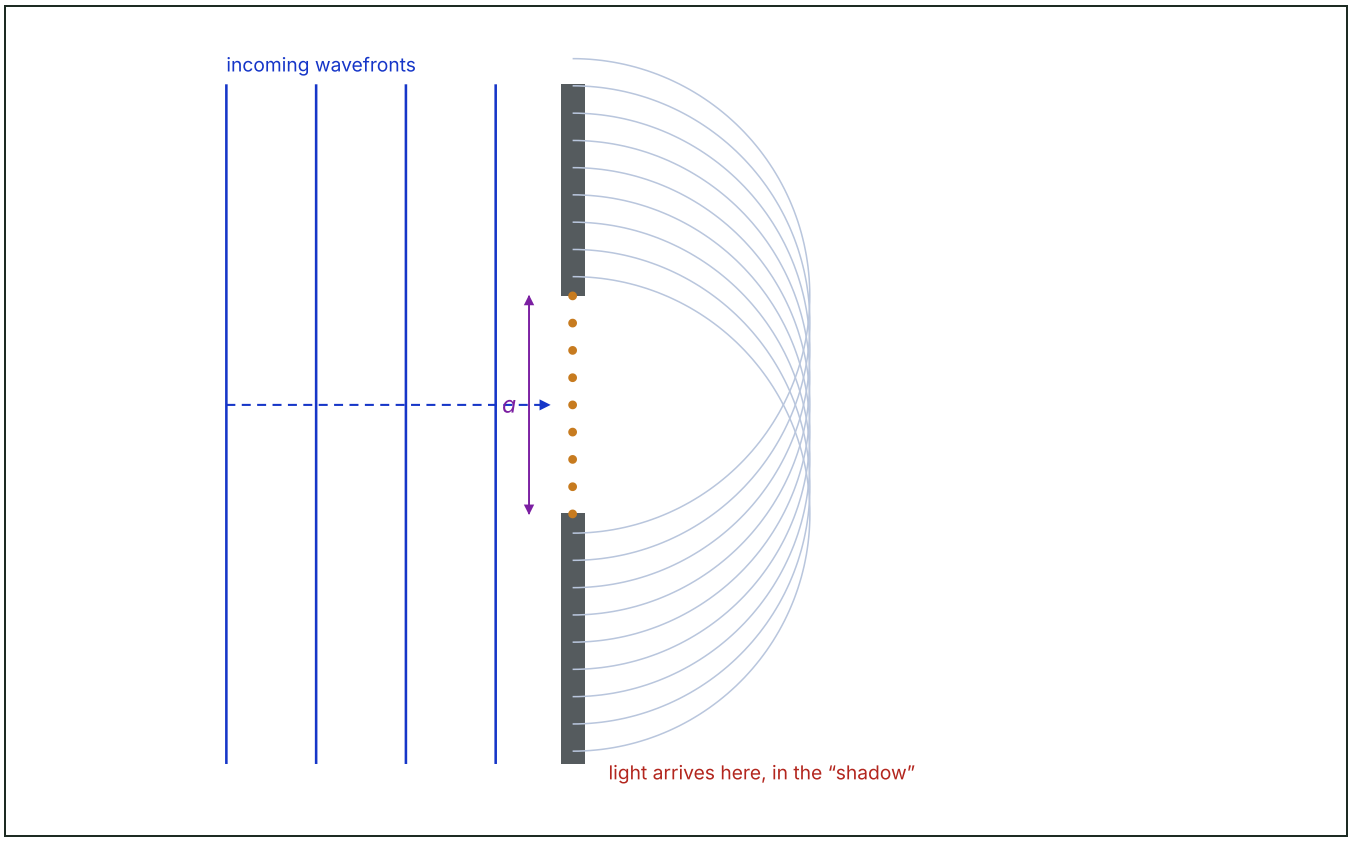



**Figure 1. Huygens’ principle.** Every point of an advancing wavefront is a new centre of disturbance from which independent waves emanate in all directions — it serves as a source of **secondary wavelets**, and the new wavefront is the **tangential surface** of all of them. In the middle the tangent surface is flat, so the wave carries straight on. **At the edges it curves round**, and that is diffraction.

**Your job on Figure 1.** The secondary wavelets are drawn for you. Construct the new wavefront — the surface that just touches all of them — and shade the region where light has reached a place the barrier should have shadowed. On Figure 2, join each ray to the partner that cancels it.

### One slit, and why it goes dark

Huygens says every point across the opening is itself a source. So the light arriving at one spot on the screen is the sum of contributions from *every* point of the slit — and those contributions can cancel each other.

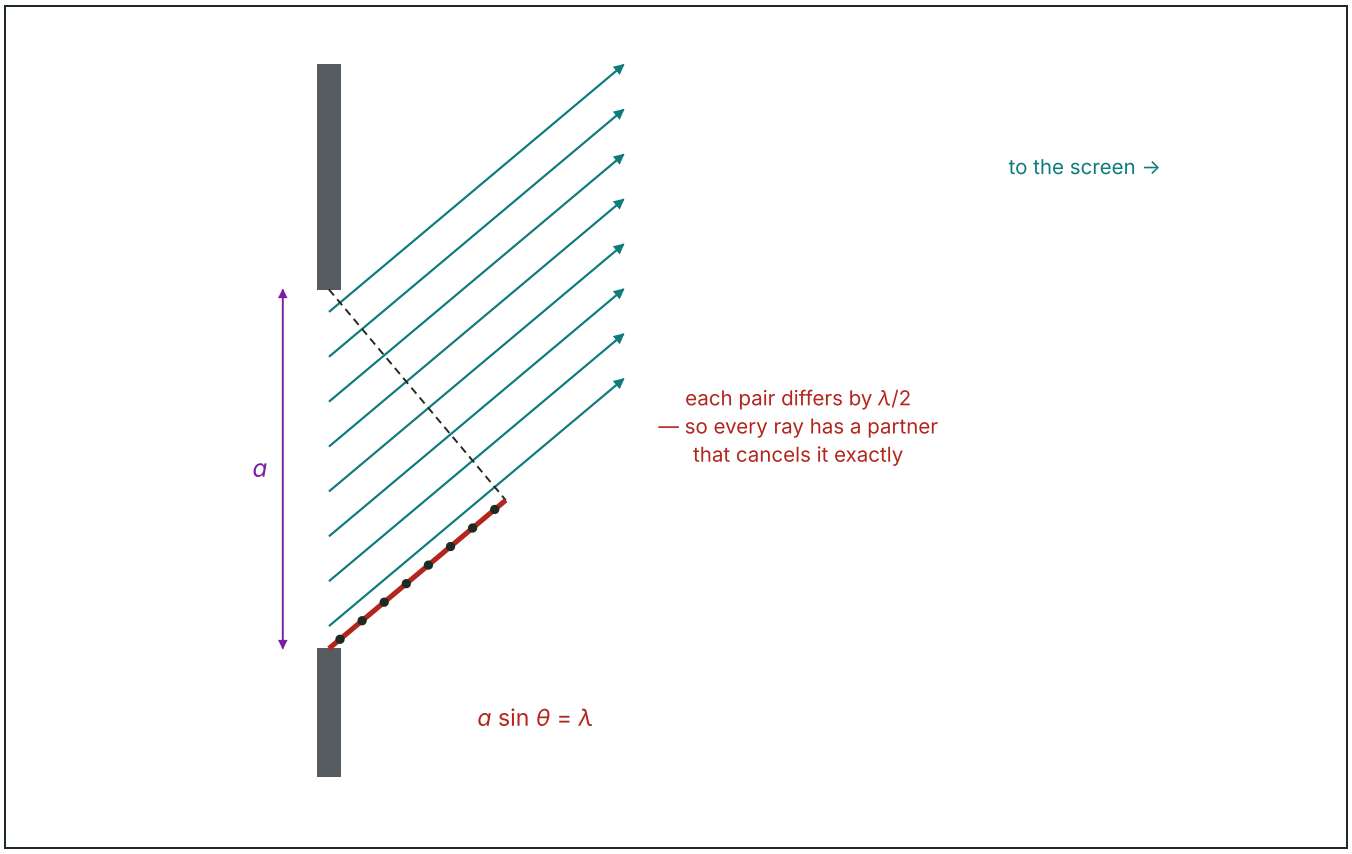



**Figure 2. Pairwise cancellation.** At the angle where $a \sin\theta = \lambda$, the rays spread evenly across the red segment. Pair each ray in the top half with the one directly below it in the bottom half: every pair is $\lambda/2$ apart, so every pair cancels, and **nothing at all** reaches the screen at that angle. That is why $a \sin\theta = m\lambda$ gives **darkness** here, and brightness in Note 10.

Walk along the four cases and the pattern appears:

| Condition | What happens across the slit | Result |
|---|---|---|
| $a \sin\theta = 0\lambda$ | every ray in step with every other | **Bright** — the central maximum |
| $a \sin\theta = 1\lambda$ | pairwise cancellation for all rays | **Dark** |
| $a \sin\theta = {}^{3}\!/_{2}\lambda$ | some rays survive and do not cancel | **Bright** — a side peak |
| $a \sin\theta = 2\lambda$ | pairwise cancellation for all rays | **Dark** |

**THE SINGLE-SLIT MINIMA**

$$a \sin\theta = m\lambda \quad \sin\theta = m\lambda/a \quad m = \pm1, \pm2, \pm3\ldots$$

and these are **DARK**. Note that $m = 0$ is excluded: the centre is the brightest point of the whole pattern.

## Worked example

A laser of $\lambda$ = **550 nm** falls on a single slit. The **second diffraction minimum** is at $\theta$ = **45°**.

1. **(1)** What is the width of the slit? **(2)** At what angle is the first minimum?

## How bright, not just where

The conditions above say where the pattern is dark. The full intensity is

$$I = I_0 \, (\sin \beta / \beta)^2, \quad \beta = \pi D \sin \theta / \lambda$$

where $D$ is the **slit width**. At $\theta = 0$ the ratio $(\sin \beta/\beta) \to 1$, so $I = I_0$: the central bright fringe. At $\beta = \pi$ and $\beta = 2\pi$, $\sin \beta = 0$ and so $I = 0$ — the dark fringes again, from the other direction.

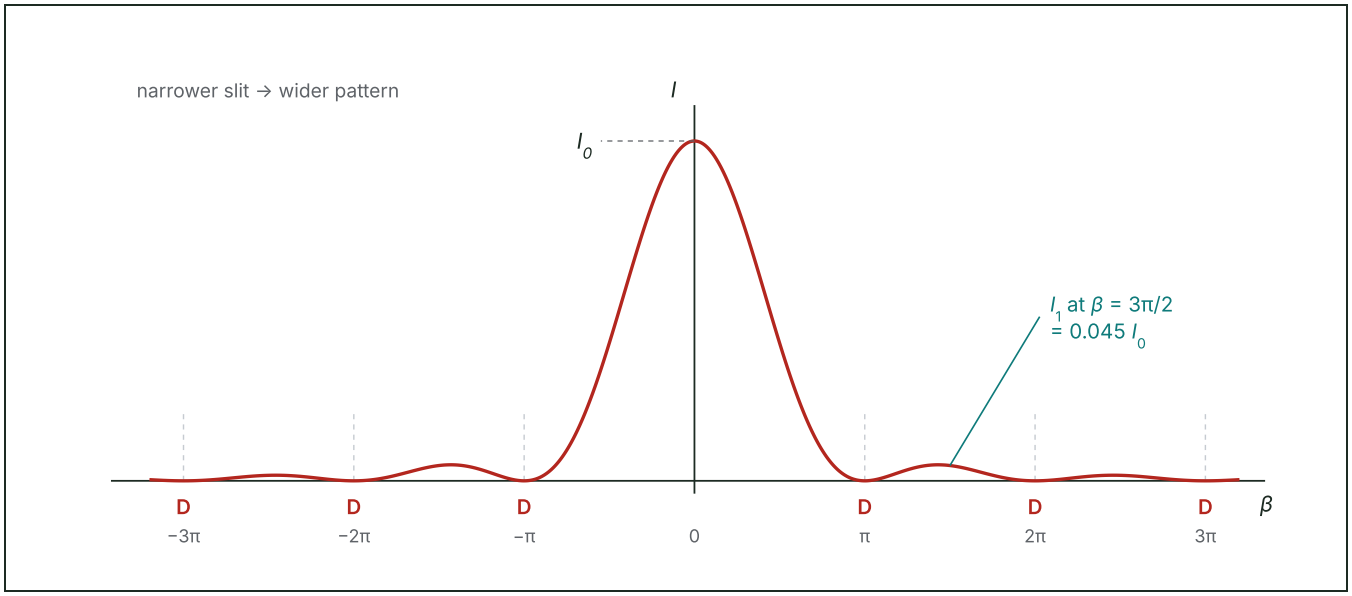



**Figure 3.** The single-slit intensity, sampled from $I = I_0(\sin \beta/\beta)^2$. **D** marks each dark fringe: the curve touches zero at every whole multiple of π. The first side peak reaches **4.5 %** of the centre — which is why the central maximum is the only part of a single-slit pattern that is easy to see.

The first side peak sits at $\beta = 3\pi/2$, midway between the first two zeros:

$I_1 = I_0(\sin(3\pi/2) / (3\pi/2))^2 = \mathbf{0.045\ I_0}$

and because $\beta$ carries $D$ in the numerator, **widening the slit narrows the pattern**. Squeeze the slit and the light spreads out further — which is the opposite of what a ray picture would predict, and is the clearest everyday sign that light is a wave.

**THE PUNCHLINE**

Two slits: $m\lambda$ is bright. One slit: $m\lambda$ is dark.

Back to the prediction: in Note 10, $d$ is the gap **between** two sources and $m\lambda$ means they arrive in step. Here, $a$ is the width of a single source, $m\lambda$ is the spread **across** it, and it means every ray finds a partner half a wavelength away. **Same algebra, opposite physics.**

## Looking ahead

Interference needed two paths through air. **Note 12** gets its two paths from a single film of soap or coating — and runs into something neither of the last two notes needed: reflection can flip a wave over.

2. Light of 600 nm passes a slit 2.0 µm wide. At what angle, in degrees, is the **first** minimum?

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3. A slit is made **narrower**. What happens to the diffraction pattern?

○ It spreads out and gets wider. ○ It narrows toward a sharp bright line. ○ It is unchanged; only the brightness falls. ○ The fringes reverse, dark for bright.

4. What fraction of the central intensity does the first side peak reach? Give it as a decimal.

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